Volume limits for long alternating-generator three-braids

1 November 2026

Abstract

Fix an even number $m \geq 6$ of syllables in a closed three-strand braid, with generators alternating between σ_2 and σ_1. For independently sufficiently long signed powers, whenever the closure is a knot, its reduced Kashaev invariant has exponential rate equal to the hyperbolic volume of the knot complement. The limit includes both parities and composite levels. We identify the original crossing disks with an explicitly matched bipyramid and compute its marked packing. Three shared choices of face weights give a uniform $7/100$ loss for every complementary source sector, for every fixed even m. The full all-size estimate and the fixed rational constant bounds are proved here. Exact finite source, local cofactor, and nonzero geometric filling results are stated explicitly as companion inputs; their analytic and geometric proofs are not reproduced in this paper.

The volume question originates with Kashaev [1] and its colored-Jones formulation with Murakami and Murakami [2]. Here every diagram and every signed power is fixed before the quantum level tends to infinity. The phrase *alternating generators* describes the braid word, not the crossing signs of its projection. Geometric volume estimates for closed three-braids appear in Futer–Kalfagianni–Purcell [3]; no geometric bound from that work is used as a substitute for the root-of-unity asymptotic below.

The proof separates its three companion inputs from its new comparison. Companion Theorem 1 identifies the original invariant with a shared coefficient and a bounded finite filling functional. Companion Theorem 2 supplies phase-uniform local rates. Companion Theorem 3 evaluates the nonzero central contribution and identifies its action with the actual filled volume. The main work here is a shared, global bound for all the other terms; this is what permits the central contribution to determine the full invariant. A bound for isolated factors with unrelated face weights would not suffice.

1 The sphere and the theorem

Throughout, fix an even integer $m \geq 6$. In the diagram source the quantum level is $N \geq 3$, and $d = N - 1$. Use poles N_0, S_0, rim vertices $v_0, \dots, v_{m-1}$, and the oriented triangles

$$(N_0, v_j, v_{j+1}), \qquad (S_0, v_{j+1}, v_j), \qquad j \in \mathbb{Z}/m. \tag{1.1}$$

Match north triangles across $N_0 v_j$ for odd j, and south triangles across $S_0 v_j$ for even j. Every triangle belongs to exactly one matched pair. This is a simple triangulated sphere with m matched pairs and $m + 2$ white vertices.

The cyclic quadrilateral order is (x, b, y, c); the homogeneous source argument order is (x, b, c, y). The original frames are

	(x, b, c, y)	matching edge
j odd	$(N_0, v_{j-1}, v_{j+1}, v_j)$	$N_0 v_j$
j even	$(S_0, v_{j+1}, v_{j-1}, v_j)$	$S_0 v_j$.

(1.2)

The reversed south middle order is essential. Mark the unmatched edge v_0v_1.

Write R for the positive half-twist of the two strands in the original disk frame; R^n is its signed integral power. Add one crossing circle around each displayed box. Removing its full twists, while retaining the chosen signed half twist, gives the augmented exterior Y_m^ϵ with these original disk markings. Retain a signed half twist $\epsilon_j \in \{-1, 0, 1\}$ at each original crossing disk. Write μ_j for its meridian and λ_j for the disk-framed primitive longitude, with the orientation convention of the original two-strand crossing. Filling kills $\mu_j + p_j\lambda_j$. In this basis, filling parameter $p_j \in \mathbb{Z}$ inserts the power

$$n_j = \epsilon_j - 2p_j. \tag{1.3}$$

Let the braid generators use the same positive crossing as R, and write factors in the order encountered by the strands.

Theorem 1. *The filled diagram is the ordinary closure of*

$$\beta_m(\mathbf{n}) = \sigma_2^{n_0}\sigma_1^{n_1}\sigma_2^{n_2}\sigma_1^{n_3}\cdots\sigma_2^{n_{m-2}}\sigma_1^{n_{m-1}}. \tag{1.4}$$

Denote this closure by $K_{m,\mathbf{n}}$. The m crossing disks are the disks surrounding those syllables. The braid axis is not a link component. The main link is a knot exactly when the permutation of (1.4) is a 3-cycle. In particular, exactly one odd north exponent and one odd south exponent, with every other exponent even, gives a knot for every m.

For each fixed m, the estimates below and the companion filling results give a finite B_m such that, whenever the fixed integers satisfy $\min_j |n_j| > B_m$ and the knot test, the closure is hyperbolic and

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N(\widehat{\beta_m(\mathbf{n})})| = \mathrm{Vol}(S^3 \setminus \widehat{\beta_m(\mathbf{n})}). \tag{1.5}$$

Here K_N is the zero-framed reduced Kashaev normalization with unknot value one. The signs and relative sizes of the fixed powers are unrestricted. There is no uniformity in growing m, no growing-twist assertion, and no moving-color conclusion.

2 Exact strand maps and the three-strand braid

Number the sides of a cyclic quadrilateral by

$$0 = xb, \quad 1 = by, \quad 2 = yc, \quad 3 = cx. \tag{2.1}$$

These are the native cut corners (C_1, C_2, C_3, C_0), respectively. The literal pure shaded map $C_1 \mapsto C_0,\ C_2 \mapsto C_3$ gives main-strand pairs $(0, 3), (1, 2)$. Either signed odd half twist gives $(0, 2), (1, 3)$. The shaded grouping of corners is a different pairing and is not used to count main components.

Embed the sphere with N_0 inside the equator and S_0 outside. The outside strips are exactly these:

- the edge v_jv_{j+1} joins boxes $j, j + 1$;
- an unmatched north spoke, at even rim index j, joins odd boxes $j - 1, j + 1$;
- an unmatched south spoke, at odd rim index j, joins even boxes $j - 1, j + 1$.

Under the pure pairing they form three nested planar circles: north, equator, south. There are no other strips. Straighten these three circles into concentric tracks while retaining the disjoint disk neighborhoods. Increasing angular coordinate is braid time; label the tracks from inside to outside by $1, 2, 3$.

The forward endpoints at each box are

	input, inner then outer	output, inner then outer
j odd	$(N_0v_{j-1}, v_{j-1}v_j)$	(N_0v_{j+1}, v_jv_{j+1})
j even	$(v_{j-1}v_j, S_0v_{j-1})$	(v_jv_{j+1}, S_0v_{j+1}).

(2.2)

North boxes therefore act on tracks $1, 2$; south boxes act on tracks $2, 3$. The unused track simply passes the box through its outside strip.

The north source frame already has increasing angular time and the inside track on its left. At a south box, both the time direction and the transverse order are reversed. Thus the change from its source frame to (2.2) is a planar half-turn. It is not a reflection or a one-head orientation reversal.

For a single crossing a planar half-turn preserves each diagonal pair of endpoints and therefore preserves which diagonal is over. It retains the signed crossing R. For R^n the order of the identical signed crossings is reversed, and the exponent is still n. Reversing both strand orientations also preserves crossing writhe. This is a literal tangle isotopy with the ribbon normal transported; it is not the assertion $T^2 = I$ for typed quantum turns.

Cut the annulus before box 0. Formula (2.2) gives (1.4) with no sign inversion at the south boxes. Closing the cut tracks gives the ordinary braid closure. The half-turn and straightening carry the original crossing circles and their disk collars along, so the identification preserves the designated twist regions rather than just the unmarked link.

Components of a braid closure are the cycles of its strand permutation. With one odd box of each kind the permutation is a product of the two distinct transpositions $(12), (23)$, hence a 3-cycle, in either order. All even exponents give three components. All odd exponents give

$$\#\text{main components} = \gcd(3, m/2). \tag{2.3}$$

These statements hold for every m by (2.2), not by a finite experiment. More general parity patterns are tested exactly by the permutation of (1.4).

The phrase "alternating generators" does not assert an alternating link diagram. For example, assigning opposite signs to the two generator types gives the usual alternating braid pattern; arbitrary signs need not.

3 Complete packing and the marked curvatures

Put

$$\theta = \pi/m, \qquad a = \csc\theta. \tag{3.1}$$

In the complex projection plane take the north disk $|z| \le a - 1$, the south **exterior** disk $|z| \ge a + 1$, and the unit disks

$$v_j : \quad |z - ae^{2ij\theta}| \le 1. \tag{3.2}$$

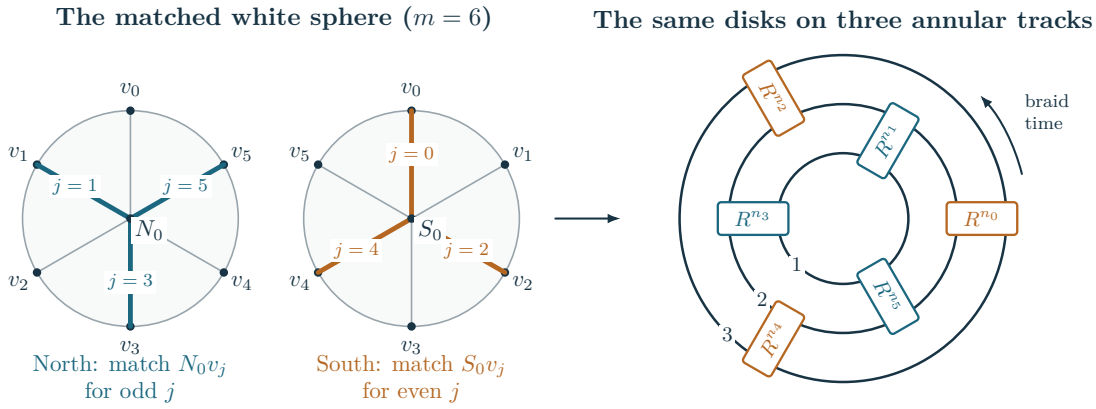
Every rim disk is tangent to both pole disks. Rim centers at cyclic separation r have distance $2a\sin(r\theta)$, which is 2 exactly for adjacent indices and strictly greater than 2 for every nonadjacent pair. The two pole disks are disjoint. Thus the contact graph, with its exterior south region, is exactly (1.1); no extra edge is introduced.

The marked contact of v_0, v_1 is

$$p = a\cos\theta\, e^{i\theta}.$$

Use the holomorphic Möbius transformation

$$w = \frac{2}{z - p}. \tag{3.3}$$



$$\beta_6 \quad = \quad \sigma_2^{n_0}\sigma_1^{n_1}\sigma_2^{n_2}\sigma_1^{n_3}\sigma_2^{n_4}\sigma_1^{n_5}.$$

Figure 1: The case $m = 6$. Equal rim labels on the two hemispheres are identified with opposite boundary orientations. A thick matching spoke pairs the two neighboring triangles and specifies one crossing disk. The right-hand picture uses those same disks: odd indices act on the inner two tracks, even indices on the outer two. Each box contains its original signed power. The picture illustrates the endpoint table (2.2), whose proof applies to every even $m \geq 6$.

A further rotation or translation puts the two protected circles into horizontal strip position without changing radii.

Under $z \mapsto 1/(z-p)$, a circle of center c and radius r has new radius $r/\big||c-p|^2 - r^2\big|$. A zero denominator gives a line. For the north and south boundaries the denominators before taking absolute value are $2(a-1)$ and $-2(a+1)$. Both circles have inversion radius $1/2$, and their oriented disks become bounded because p is outside both original disks. The factor 2 in (3.3) gives both new radii one.

For rim circle j,

$$\begin{aligned} |ae^{2ij\theta} - p|^2 - 1 &= 2a^2\cos\theta\{\cos\theta - \cos((2j-1)\theta)\} \\ &= 4a^2\cos\theta\sin(j\theta)\sin((j-1)\theta). \end{aligned}$$

It vanishes exactly at $j = 0, 1$ and is positive elsewhere. Therefore the complete marked curvatures, meaning reciprocals of Euclidean circle radii, are

$$L_{N_0} = L_{S_0} = 1, \qquad k_j = L_{v_j} = \frac{\cos\theta}{\sin^2\theta}\{\cos\theta - \cos((2j-1)\theta)\}. \tag{3.4}$$

In particular,

$$k_0 = k_1 = 0, \quad k_j > 0 \ (2 \leq j \leq m-1), \quad k_2 = k_{m-1} = 4\cos^2\theta. \tag{3.5}$$

The two protected weights are exact zeros. They are never replaced by small positive curvatures.

The strip width is 2, since the radius-one north circle is tangent to both parallel lines. If another source convention uses width 1, a common rescaling multiplies all ordinary curvatures by 2. The action is unchanged by common scaling because total numerator degree equals total ordinary target. Formula (3.4) is the normalization with pole curvatures one.

The complete-geometry assertion used below has two ingredients. First, the simple-sphere construction of [11, Sections 2–3] turns this packing into the labeled right-angled ideal polyhedron: white faces correspond to the displayed circles and shaded triangles to

the interstices. Its hypotheses hold because the contact graph just computed is the simple triangulated sphere (1.1). Second, the original cut is determined by the matching. Cutting the annular diagram along the projection sphere and the original crossing half disks gives two balls with these labeled faces. White faces pair across the projection copies; shaded faces pair according to (1.2), with the retained half twist exchanging their copies and punctures when prescribed. These are precisely the inverse-cut identifications of [13, Section 2]. Thus that construction applies to Y_m^ϵ, with its original disks and main strands. No change to a different shaded-face matching is made.

4 The exact source and what must be bounded

Companion theorem 1 (Exact source and finite filling norm)**.** *For the original cut of a fixed connected marked simple matched sphere at $N \geq 3$, let $d = N - 1$, let Ω be the complete-color two-strand Casimir, and form its polynomial numerator $V_N(\sigma;\Omega) = (2T_N(\sigma/2) + 2)(\sigma - \Omega)^{-1}$. The homogeneous face polynomial $H_V(x, b, c, y; \sigma)$ is the Fourier transform of this original matrix numerator in the four boundary states, with the turn and crossing frames retained. It has total face degree d and spectral degree at most d. Here $T_N(\cos\phi) = \cos(N\phi)$. The full diagram source is its shared coefficient with target d at every ordinary face and target zero at the two faces incident to the marked strand edge. The raw filled evaluation is the tensor product of the original signed-power functionals $\mathcal{L}_N^{n_e}$ applied to that polynomial, and differs from the zero-framed reduced invariant only by its prescribed unit-modulus character. For every complex polynomial H of degree at most d,*

$$|\mathcal{L}_N^n H| \leq (1 + |n|/2) \max_{-2\leq\sigma\leq 2} |H(\sigma)|, \qquad n \in \mathbb{Z}.$$

The scalar polynomial norm also holds at $N = 2$; the diagram source is used only at $N \geq 3$. These statements hold for both level parities, with first normal data at every double spectral root and the value at the odd simple root.

This is a companion result, not a theorem proved anew here. The exact Morse coefficient and its face targets are [5, Sections 4–6]; the matrix-to-face adapter and scalar norm are proved in [7, Theorem 1.1 and Sections 2–7], with the original diagram normalization recorded separately in [6, Section 1]. The finite functional theorem does not itself identify a knot diagram with a shared coefficient. That identity is the separately cited Morse source result. These inputs are also recorded in Appendix N of [4]. Their hypotheses will be checked on the explicit sphere below. The local decomposition into a central term and the two one-sided recurrence pieces is the one of [8, Sections 1–3]; no positive replacement of the full matrix numerator is made.

Put $q = e^{\pi i/N}$, $d = N - 1$, and

$$\chi_N(z) = 2T_N(z/2) + 2, \qquad V_N(z;\Omega) = \chi_N(z)(z - \Omega)^{-1}. \tag{4.1}$$

Polynomial division defines the degree-at-most-d matrix polynomial in (4.1), including at every spectral root. It is the original Casimir numerator, not the adjugate of an unrelated matrix. Let $H_V(x, b, c, y; z)$ be its original homogeneous four-face polynomial.

Writing the pole variables as A, B and the rim variables as z_j, the full shared spectral polynomial is exactly

$$S_N(\boldsymbol{\sigma}) = [A^d B^d \prod_{j=2}^{m-1} z_j^d] \prod_{j\,\text{odd}} H_V(A, z_{j-1}, z_{j+1}, z_j; \sigma_j) \prod_{j\,\text{even}} H_V(B, z_{j+1}, z_{j-1}, z_j; \sigma_j)\Big|_{z_0=z_1=0}. \tag{4.2}$$

Every local factor has total face degree d and spectral degree at most d. To define its sectors, put $h = \lfloor N/2 \rfloor$. An original decreasing matrix path from row j to row i has middle monomial $b^i c^{d-j}$. At even $N = 2h$, it is central when $i \le h-1$ and $j \ge h$; at odd $N = 2h+1$, it is central when $i \le h$ and $j \ge h$. The left sector has $j < h$, and the right sector consists of the remaining paths entirely above the central cut. These three sets partition the original numerator paths. Let $S_{N,\mathrm{C}}$ select the central part at every box by polynomial linearity, and set

$$D_N = S_N - S_{N,\mathrm{C}}. \tag{4.3}$$

The task is a strict real-cube bound on (4.3). The complete drilled rate of S_N alone would not give the smaller fixed-filled rate.

The general Morse compiler proves (4.2) on this diagram. In its proof each actual box has factor $N^{-1} H_V/(bc)^{d/2}$; the physical cups, caps and unrestricted mark leave N^{-m-1}. The primitive face-potential lattice has $m+1$ independent N-point grids, canceling that factor. Endpoint support is imposed before specializing those grids. The total numerator degree md forces the unique nonnegative alias: the two marked targets are zero and every other target is d. This retains all repeated-node normal channels.

At even levels a literal integrated parity cochain is

$$t(v_j) = j \pmod 2, \qquad t(N_0) = 1, \qquad t(S_0) = 0. \tag{4.4}$$

It agrees across matched spokes and differs across each strand edge; in particular the two protected endpoints differ. The integrated coset character at the unique supported target is one. No extra parity or power of N multiplies (4.2).

The raw filled contraction is

$$Z_N(\mathbf{n}) = \left(\bigotimes_{j=0}^{m-1} \mathcal{L}_N^{n_j} \right) S_N, \tag{4.5}$$

where $\mathcal{L}_N^n$ is the original full-Hermite functional for the signed power R^n. Its crossing scalar and all first derivatives at double roots are part of its definition. Conversion from (4.5) to the specified zero-framed reduced invariant is the original unit-modulus complete-color framing/component character. The separate full one-cut modified convention retains i^{N+3}/N; it also has zero exponential rate.

These exact facts use [5], §§5–6, and [6], §1. They use the displayed Morse diagram itself, not an assumed flat-plat presentation or a scalar tree contraction.

5 The spectator envelope is exactly the union of the genuine rates

Use the source-normalized triangle potential

$$\begin{aligned} Q(w_1, w_2, w_3) &= \max_{\substack{\alpha_i \ge 0 \\ \sum_i \alpha_i = 1/2}} \left\{ \frac{1}{\pi} \sum_i [\Lambda(\pi\alpha_i) + \Lambda(\pi/2 - \pi\alpha_i)] + \sum_i \alpha_i \log w_i \right\}, \\ \Lambda(t) &= -\int_0^t \log|2\sin u|\, du. \end{aligned} \tag{5.1}$$

At a zero weight restrict its load to zero and omit its logarithmic term from the sum. For one positive weight it is one half of its logarithm. It is symmetric, nondecreasing, and obeys

$$Q(cw) = Q(w) + \tfrac{1}{2}\log c, \qquad Q(x,t,y) \ge \tfrac{1}{2}\log t \quad (t > 0). \tag{5.2}$$

The second inequality follows by using the admissible vertex with load $1/2$ on t. Its entropy is zero since $\Lambda(0) = \Lambda(\pi/2) = 0$. This proves the bound also when either other coordinate is protected.

For a fixed nonnegative face vector with $x + y > 0$, the genuine uniform local upper rates are

$$\begin{aligned} X &= Q_b + Q_c, \qquad Q_b = Q(x, b, y), \quad Q_c = Q(x, c, y), \\ A_{\mathrm{L}} &= \max(\log c, Q_b + \tfrac{1}{2}\log c) \quad (c > 0), \\ A_{\mathrm{R}} &= \max(\log b, Q_c + \tfrac{1}{2}\log b) \quad (b > 0). \end{aligned} \tag{5.3}$$

An overloaded zero middle gives the zero one-sided polynomial and is omitted. From (5.2),

$$\begin{gathered} Q_b + \tfrac{1}{2}\log c \le X, \qquad Q_c + \tfrac{1}{2}\log b \le X, \\ \max(X, A_{\mathrm{L}}, A_{\mathrm{R}}) = H := \max(X, \{\log b : b > 0\}, \{\log c : c > 0\}). \end{gathered} \tag{5.4}$$

Thus the spectator H used by the three-vector proof is precisely the maximum of the genuine local rate bounds. No mixed term has been lost. This identity is already recorded in [10], equation (4), and in the final paragraph of the global LEFT theorem.

Companion theorem 2 (Uniform local sector bounds)**.** *Fix nonnegative face moduli (x, b, c, y) with $x + y > 0$, keeping each protected modulus exactly zero. Write $H_{V,s}$ for the original central or one-sided polynomial, where $s \in \{\mathrm{C}, \mathrm{L}, \mathrm{R}\}$. On the face torus with these moduli, its real-spectral upper exponential rate is at most $X, A_{\mathrm{L}}, A_{\mathrm{R}}$, respectively, as defined in (5.3). The bound is uniform on the entire spectral interval $[-2, 2]$, includes both parities of N, and does not depend on the face phases. A one-sided term whose overloaded middle has modulus zero is identically zero.*

The proof is imported from [8, Section 8] and [9, Sections 1–4]. The following explanation records which uniformity is required; the local recurrence and its global transfer estimate are not being replaced by a positivity assumption. Here "uniform local upper rate" means

$$\limsup_{N\to\infty} \frac{1}{N} \log \sup_{\substack{\sigma\in[-2,2] \\ |L_f|=\rho_f}} |H_{V,s}(L;\sigma)| \le A_s(\rho), \qquad s = \mathrm{C}, \mathrm{L}, \mathrm{R}. \tag{5.5}$$

For CROSS this follows from the cleared-cofactor coefficientwise real-interval interpolation bound and the positive complete triangle factors. For LEFT the original omitted-half product and both unused spectral products remain. The normalized interval continuant has nonnegative face coefficients on $[-2, 2\cos(\pi/N)]$; taking face moduli therefore preserves the signed-continuant upper action. Its complete endpoint optimization gives (5.3). The remaining real cap has width $O(N^{-2})$. Since the actual spectral degree is at most d, polynomial exterior comparison costs only

$$\exp\left[d \operatorname{arcosh} \frac{6 - 2\cos(\pi/N)}{2 + 2\cos(\pi/N)}\right] = O(1). \tag{5.6}$$

This applies to a polynomial with complex coefficients at each fixed face-phase tuple. RIGHT follows by the actual row/middle reversal, with b, c exchanged in its rate. These are the proofs of [9], §§1–4, and [8], §8. They include both parity conventions and sampled spectral zeros. A polynomial coefficientwise LEFT bound against its complete value is neither asserted nor needed.

In this family the outer pole is always positive, so $x + y > 0$ at every box and every one of the three proposed vectors. The zero-coordinate limits in (5.1)–(5.5) are therefore within the proved domain.

6 Complete packing and the three fixed shared vectors

Let $\theta = \pi/m$. The complete packing has concentric pole boundaries of radii $a_0 - 1, a_0 + 1$, where $a_0 = \csc\theta$, and unit rim disks centered at $a_0 e^{2ij\theta}$. The south disk is the exterior of its boundary. Adjacent rim disks are tangent and all other rim pairs are disjoint. Invert at the marked rim contact and multiply lengths by two. Both poles then have curvature one, the two protected faces become lines, and the remaining curvatures are

$$k_j = \frac{\cos\theta}{\sin^2\theta}[\cos\theta - \cos((2j-1)\theta)] = \frac{2\cos\theta}{\sin^2\theta}\sin(j\theta)\sin((j-1)\theta). \tag{6.1}$$

Thus $k_0 = k_1 = 0$, all other $k_j > 0$, and $k_2 = k_{m-1} = 4\cos^2\theta$.

The labeled simple-sphere packing and its positive cut polyhedron identify the original augmented complement. The calibrated complete CROSS action is

$$\Phi_m = 2\sum_{j=0}^{m-1} Q(1, k_j, k_{j+1}) - \sum_{j=2}^{m-1} \log k_j = \frac{\mathrm{Vol}(Y_m^\epsilon)}{2\pi}. \tag{6.2}$$

The equality uses the original marked triangle calibration. It is not a full positive-endpoint source theorem. Deleting the two protected faces leaves a connected ordinary graph, since both poles meet every vertex of the surviving rim path. Thus its face Hessian has only the common-scale kernel, as required by the Morse CROSS theorem. Every matched edge has a positive outer and at least one nonmonomial triangle.

Put

$$a = \frac{9}{4}, \quad C = \frac{27}{4}, \quad \lambda = \frac{a}{k_2} = \frac{9}{16\cos^2\theta}, \quad w_j = \lambda k_j, \qquad L_{N_0} = L_{S_0} = 1. \tag{6.3}$$

The whole vector W has $w_2 = w_{m-1} = a$ and

$$c := w_3 = w_{m-2} = a(4\cos^2\theta - 1) \in [9/2, C]. \tag{6.4}$$

Define $W^{(1)}$ by replacing only w_2 with 1, and $W^{(0)}$ by replacing only w_{m-1} with 1. Each shared occurrence of that face is changed at once. Protected weights remain exactly zero.

The surviving one-sided anchors at $j = 0, 1$ are LEFT; those at $j = 2, m-1$ are RIGHT. Every other box has both possible one-sided anchors. This follows directly from (1.2), not from a reflected physical coupon. Use $W^{(1)}$ for 1L, $W^{(0)}$ for 0L, and W for all other anchors.

7 Bounds for every even number of syllables

Write $F(r) = Q(1, r, 0)$. Scale only the ordinary rim coordinates by $t \in (0, 1]$, retaining pole weights one, and let $P(t)$ denote the complete action at that vector. The pole angle in triangle j is

$$\alpha_j(t) = \arctan\frac{1}{\sqrt{t(k_j + k_{j+1}) + t^2 k_j k_{j+1}}}, \qquad \alpha_0(t) = \frac{\pi}{2}. \tag{7.1}$$

The complete star gives $\sum_j \alpha_j(1) = \pi$. For every other triangle, with $D^2 = t(u+v) + t^2uv$,

$$-\frac{d\alpha}{d\log t} = \frac{t(u+v) + 2t^2uv}{2D(1+D^2)} \le \sin\alpha\cos\alpha \le \alpha. \tag{7.2}$$

Hence $\alpha_j(t) \le \alpha_j(1)/t$, and differentiation of the entire coupled action gives

$$\frac{dP}{d\log t} = 2 - \frac{2}{\pi}\sum_j \alpha_j(t) \ge 1 - \frac{1}{t}.$$

Integrating in the decreasing direction yields

$$P(\lambda) - \Phi_m \le \lambda^{-1} - 1 + \log\lambda \le D_0 := \frac{16}{9} - 1 + \log\frac{9}{16} < \frac{203}{1000}. \tag{7.3}$$

The bound is independent of the number of triangles; it uses their original complete star, rather than summing independent local distortions.

For $3 \le j \le m-2$, (6.1) gives

$$\frac{1}{3} \le \frac{w_{j-1}}{w_j} \le 2, \qquad \frac{1}{3} \le \frac{w_{j+1}}{w_j} \le 2. \tag{7.4}$$

The first ratio is $\sin((j-2)\theta)/\sin(j\theta)$, increasing with j; the second is $\sin((j+1)\theta)/\sin((j-1)\theta)$, decreasing. Their extremal bounds use $\sin 3\theta \le 3\sin\theta$ and $\sin 4\theta/\sin 2\theta = 2\cos 2\theta \le 2$, valid also at $m = 6$.

The logarithmic derivative of Q in a positive weight lies in $[0, 1/2]$. Therefore F is increasing and $F(r) - \log r$, $F(r) - \frac{1}{2}\log r$ are decreasing. Drop the positive pole, scale by the outer rim coordinate, and use (7.4). At every interior box,

$$\begin{aligned} X_j - \log\max(w_{j-1}, w_{j+1}) &\ge F(1/3) + F(2) - \log 2 =: B_{\text{int}} > 43/100, \\ Q(1, w_{j\pm1}, w_j) - \tfrac{1}{2}\log w_{j\pm1} &\ge F(2) - \tfrac{1}{2}\log 2 =: C_{\text{int}} > 21/50. \end{aligned} \tag{7.5}$$

At boxes 2 and $m-1$, the middle weights are $0, c$ and the outer rim weight is a. The same derivative bounds and $c \le C$ give

$$\begin{aligned} X_j - \log c &\ge F(a) + Q(1, C, a) - \log C =: B_{\text{near}} > 7/25, \\ Q(1, c, a) - \tfrac{1}{2}\log c &\ge Q(1, C, a) - \tfrac{1}{2}\log C =: C_{\text{near}} > 21/50. \end{aligned} \tag{7.6}$$

At boxes 0 and 1, $X = F(a) > \log a$. Thus every spectator has $H = X$ at W. An anchored LEFT loss is $\min(X - \log c, Q_c - \frac{1}{2}\log c)$, and the RIGHT formula is symmetric. Equations (7.5)–(7.6), less (7.3), give a net loss greater than $7/100$ for every anchor other than 0L, 1L. In particular even the weakest coarse comparison gives $7/25 - 203/1000 = 77/1000 > 7/100$.

For 1L, use $W^{(1)}$. Its actual anchored rate is now zero and its complete rate is $F(1)$. The entire complete-action change from W is

$$T(c) = 2[F(1) - F(a) + Q(1, 1, c) - Q(1, a, c)] + \log a. \tag{7.7}$$

It increases in c: the logarithmic derivative at c of the first Q is larger than that of the second. The only possible spectator excess is at box 2,

$$E(c) = \max(0, \log c - F(1) - Q(1, 1, c)) \le E(C). \tag{7.8}$$

At box 3, put $z = w_4$. Here $z \ge 1$, $z/c \le 2$, and $1/c \ge 4/27$, so

$$X_3 - \log\max(1, z) \ge F(4/27) + F(2) - \log 2 > 8/25. \tag{7.9}$$

Every other box is unchanged. Therefore the anchored cost at $W^{(1)}$ is at most

$$P(\lambda) + T(C) + E(C) - F(1), \qquad S_0 := F(1) - T(C) - E(C) > 37/100. \tag{7.10}$$

Its saving beyond Φ_m exceeds $37/100 - 203/1000 = 167/1000$. Reflection of the scalar weight formulas by $j \mapsto 1-j$, accompanied by exchanging pole labels, proves the same estimate for 0L at $W^{(0)}$. It does not change the ordered quantum data in (1.2).

The fixed constants in these comparisons have the rational enclosures proved in Appendix A. The proof uses only finitely many fixed analytic constants; the estimates above hold for every even $m \ge 6$.

We have proved, at every surviving anchor (j, s),

$$A_{j,s}(L^{j,s}) + \sum_{i \ne j} H_i(L^{j,s}) - \sum_{f \text{ ordinary}} \log L_f^{j,s} < \Phi_m - 7/100. \tag{7.11}$$

8 From the three vectors to the full shared remainder

Expand (4.3) into genuine label strings in $\{\mathrm{C},\mathrm{L},\mathrm{R}\}^m$, omitting zero strings and the all-CROSS string. Each remaining string contains a surviving one-sided anchor. Assign it to one such anchor, and use that anchor's single whole face vector from Section 6.

For fixed real spectra, apply the multivariate Cauchy coefficient estimate to (4.2) on the ordinary face torus of that vector. The protected variables are already zero and have no integration circle. The local estimates (5.5), and the exact spectator identity (5.4), give

$$\frac{1}{N}\log\|D_{N,\mathrm{string}}\|_{[-2,2]^m} \le A_{j,s} + \sum_{i\neq j} H_i - \frac{d}{N}\sum_{f\ \mathrm{ordinary}} \log L_f + o_m(1). \tag{8.1}$$

The same face is used everywhere with its same weight. There is no independent local optimization, no exchange of a maximum and an infimum, and no relaxation of the shared demands.

The target difference $d/N - 1 = -1/N$ contributes only the fixed number $N^{-1}\sum_f \log L_f$. For fixed m there are finitely many vectors and at most $3^m - 1$ strings. Thus summing their absolute values costs $\log(3^m-1)/N = o_m(1)$, not an extra exponential rate. Equation (7.11) proves

$$\limsup_{N\to\infty} \frac{1}{N}\log\|D_N\|_{[-2,2]^m} \le \Phi_m - 7/100. \tag{8.2}$$

In particular, for every $0 < \eta < 7/100$, the norm is at most $\exp[N(\Phi_m - \eta) + o_m(N)]$. This statement includes even and odd levels. It requires no phase positivity of the shared source and no strict gap for the all-CROSS term itself on this whole spectral cube.

9 Full Hermite insertion introduces no new exponential cost

For H of degree at most d, the exact finite filling theorem states

$$|\mathcal{L}_N^n H| \le (1 + |n|/2)\|H\|_{[-2,2]}. \tag{9.1}$$

Its proof is the original midpoint/full-Hermite formula with the physical crossing multiplier, followed by sharp trigonometric Bernstein. It retains value and derivative terms at double roots and the odd simple endpoint. It applies to negative as well as positive powers, without a square-root normalization or omitted Cartan factor.

Both D_N and $S_{N,\mathrm{C}}$ have degree at most d in each spectral variable. Applying (9.1) successively yields

$$|Z_N - Z_{N,\mathrm{C}}| \le \prod_{j=0}^{m-1} (1 + |n_j|/2)\,\|D_N\|_{[-2,2]^m}. \tag{9.2}$$

The polynomial need not factor and its shared targets are unchanged. The stronger bound exploiting product-node vanishing is optional. For each fixed tuple $\mathbf{n}$, the product in (9.2) is independent of N; hence (8.2) is the actual filled scalar remainder rate. No Fourier-tail argument is needed for this remainder.

10 The nonzero original Morse contribution and its filled volume

Companion theorem 3 (The nonzero geometric filling contribution)**.** *Consider the original general Morse diagram of a fixed connected simple matched sphere, with its cyclic orders, crossing-disk frames and marked strand edge. Assume its complete labeled packing is realized*

by the original positive ideal-cell cut, the ordinary face graph is connected, and each matched pair has a positive outer coordinate and at least one nonmonomial triangle. Retain the original coefficient targets, integer face lattice and full spectral numerator. Fix retained half twists ϵ_e, and fill the original disks by $n_e = \epsilon_e - 2p_e$. If the main strands form one knot, then for all independently sufficiently large $|p_e|$, the actual all-central period has the nonzero asymptotic (10.1). Its selected action obeys (10.2)–(10.4); in particular it is the volume of the smoothly filled original knot exterior divided by 2π, and tends to the complete augmented action as all filling lengths tend to infinity. The main cusp remains complete. There is no condition on the relative sizes or signs of the fixed filling integers.

The analytic conclusion is [12, Sections 1–8]; the original topology, marked primitive words, smooth completion and absolute volume calibration are [13, Sections 1–9], reproduced as the two parts of Appendix O in [4]. These are substantial companion proofs. They give the central contribution, not the full invariant. The missing complementary bound is proved in this paper. The complete simple-sphere geometry used in their hypotheses is constructed in [11, Sections 2–3]; the packing below also provides its explicit circle configuration.

The simple sphere, positive labeled packing, connected ordinary graph, primitive rank-$m-1$ face quotient, and nonmonomial triangle at each matching edge have been checked in Sections 1–3 and 6. These are precisely the hypotheses of [12] and [13]. They concern the original all-CROSS projection (4.2); they do not require complete-radius balance, a middle forest, a flat sweep or a full positive-endpoint source theorem.

Choose retained $\epsilon_j \in \{-1, 0, 1\}$ with the prescribed parities and write $n_j = \epsilon_j - 2p_j$. Let $A_N(\delta)$ be the finite face-critical action in the companion period theorem: the shared face variables have been evaluated at their selected stationary point, with the original scalar removed as specified in that theorem. Define

$$\mathcal{S}_{N,\mathbf{1}}(\delta) = A_N(\delta) + \pi i \sum_j n_j \delta_j^2 + 2\pi i \sum_j \delta_j.$$

Let $\delta_{N,\mathbf{1}}$ be its selected critical point and $S_{N,\mathbf{1}}$ its critical value. Write $H_N(\delta)$ for the face Hessian on the primitive common-scale quotient, and $Q_{N,\mathbf{1}} = -\partial_\delta^2 \mathcal{S}_{N,\mathbf{1}}(\delta_{N,\mathbf{1}})$ for the negative spectral Hessian. The face square root continues from the positive complete Hessian; the spectral square root is its Gaussian branch. The raw period has the nonzero asymptotic

$$\begin{aligned} Z_{N,\mathrm{C}} = {} & q^{d^2 \sum_j n_j/4} (-2i)^m \sqrt{2\pi}\, N^{m+1/2} \\ & \times \frac{\prod_j \sin(2\pi \delta_{N,\mathbf{1},j})}{\sqrt{\det H_N(\delta_{N,\mathbf{1}})}\sqrt{\det Q_{N,\mathbf{1}}}} e^{N S_{N,\mathbf{1}}} \left(1 + O_{m,\mathbf{n}}(N^{-1}) + O_{m,\mathbf{n}}(e^{-c_{m,\mathbf{n}} N})\right). \end{aligned} \tag{10.1}$$

The companion theorem proves $A_N \to A_0$ with every fixed derivative, $S_{N,\mathbf{1}} \to S_{\mathbf{1}}$, and convergence of the coefficient remaining after the explicit action and level factors are removed to a nonzero number. In particular that coefficient is subexponential. The sine factors and determinants do not vanish at the chosen fixed long filling. The first reflected charges have already been paired by the exact finite trace; nonzero amplitude does not follow merely from a formal even phase. The source factor $(-1)^{dm}$ and Fourier factor $(-1)^{(N+1)m}$ cancel exactly. Both level parities therefore have this same nonzero normalization.

The limiting pure action A_0 has $A_0(0) = \Phi_m$, and the selected first-charge action is

$$S_{\mathbf{1}} = A_0(\boldsymbol{\delta}) + \pi i \sum_j n_j \delta_j^2 + 2\pi i \sum_j \delta_j. \tag{10.2}$$

Let $R_{0,j}$ be the similarity of the pure ordered shaded-side map before inserting the retained half twist. The original meridian threading follows its inverse, so its pure logarithm is $-\operatorname{Log} R_{0,j}$,

as in [13, Section 5, equations (14)–(16)]. Let U_j, V_j be the continuous logarithmic similarities of μ_j, λ_j, with their zero values at the complete augmented structure fixing the branches. The actual primitive logs satisfy

$$U_j = -\operatorname{Log} R_{0,j} + 2\pi i \epsilon_j \delta_j, \quad V_j = -4\pi i \delta_j, \quad U_j + p_j V_j = -2\pi i. \tag{10.3}$$

They are original disk words with the signed half-twist correction once. The source differentiation and the marked regulator identity cancel the peripheral boundary term in (10.2). The nearby framed representation, actual logarithmic link periods and the primitive equation in (10.3) give a smooth filling of those very circles; the main cusp remains complete. Consequently

$$2\pi \operatorname{Re} S_{\mathbf{1}} = \operatorname{Vol}(S^3 \setminus K_{m,\mathbf{n}}), \qquad \operatorname{Re} S_{\mathbf{1}} \longrightarrow \Phi_m \quad \text{as all } |n_j| \to \infty. \tag{10.4}$$

At the selected charge $\operatorname{Im} \delta_j > 0$, so $\operatorname{Re} V_j > 0$. The auxiliary longitude therefore has nonunit modulus. The small logarithmic periods of its two primitive generators are fixed before multiplying one by p_j. Thus (10.3) is the actual angular period $-2\pi i$, not only an equation after exponentiation. The companion metric completion fills these original solid tori and keeps the main cusp complete. Its absolute volume comparison uses the same ideal-cell chain and a boundary homotopy whose volume tends to zero at the cusp cutoff. This is the metric statement used in (10.4); a regulator identity alone would not supply it. The continuation permits independently long signed powers, with no ratio condition.

Fix $0 < \eta < 7/100$. Increase the long-filling threshold until both Morse theorems apply and the action drop is less than $\eta/2$. Equations (8.2), (9.2) and the nonzero asymptotic (10.1) give

$$\frac{Z_N - Z_{N,\mathrm{C}}}{Z_{N,\mathrm{C}}} = O_{m,\mathbf{n}}(e^{-\kappa_{m,\mathbf{n}} N}) \quad \text{for some } \kappa_{m,\mathbf{n}} > 0. \tag{10.5}$$

The unit-modulus framing character and the optional single modified factor have no exponential rate. This proves (1.5).

There are finitely many retained half-twist patterns for fixed m. Taking the maximum of their thresholds gives B_m for every knot-producing parity, not only the illustrative pattern with two odd boxes. Any signed odd n_j may, for example, be represented with $\epsilon_j = 1$ and integer $p_j = (1 - n_j)/2$; positivity of n_j is not presumed.

A Exact enclosures for the fixed constants

The inequalities in the proof depend on fixed arguments of Q, not on a sampled list of diagram sizes. For positive weights on their support, put $D^2 = \sum_{i<j} w_i w_j$ and $a_i = \arctan(w_i/D)$. Strict concavity of the triangle entropy, or direct differentiation with its sum constraint, gives

$$Q(w) = \frac{1}{\pi} \sum_{w_i > 0} \{\Lambda(a_i) + \Lambda(\pi/2 - a_i) + a_i \log w_i\}. \tag{A.1}$$

Indeed $\sum a_i = \pi/2$, and these are the stationary loads $\alpha_i = a_i/\pi$ in (5.1). When the support consists of one weight, use $Q(w) = \frac{1}{2} \log w$ directly. All arguments of Q in the following table are rational. The symbols $a, C, D_0, B_{\mathrm{int}}, C_{\mathrm{int}}, B_{\mathrm{near}}, C_{\mathrm{near}}, T, E, S_0$ are those in (6.3)–(7.10).

Fixed quantity	Rational enclosure sufficient for the proof
D_0	$D_0 < 203/1000$
$F(a) - \log a$	> 0
B_{int}	$> 43/100$
C_{int}	$> 21/50$
B_{near}	$> 7/25$
C_{near}	$> 21/50$
$F(4/27) + F(2) - \log 2$	$> 8/25$
$S_0 = F(1) - T(C) - E(C)$	$> 37/100$

Here is the enclosing arithmetic used to establish the table. Write $J(t) = \int_0^t \log(2 \sin \pi u)\, du$. For $0 < t < 1/2$, the Euler sine product, followed by termwise integration of its positive absolutely convergent series, gives

$$J(t) = t \log(2\pi t) - t - \sum_{n=1}^{20} \frac{\zeta(2n) t^{2n+1}}{n(2n+1)} + R(t), \quad -\frac{2t^{43}}{21 \cdot 43(1-t^2)} \le R(t) \le 0. \tag{A.2}$$

The tail follows from $\zeta(2n) < 2$ and $n(2n+1) \ge 21 \cdot 43$ for $n \ge 21$. The identity $\Lambda(\pi t)/\pi = -J(t)$ then evaluates (A.1) without quadrature.

Every interval endpoint is an integer multiple of 10^{-40}. Addition is exact at this scale; products and quotients take the minimum and maximum endpoint combinations and round outwards. All divisors are first checked to exclude zero. Square roots use integer floor and ceiling square roots. Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ is enclosed by forty terms of each alternating series and its next-term error. For $n \ge 2$,

$$\sum_{k=1}^{64} k^{-2n} \le \zeta(2n) \le \sum_{k=1}^{64} k^{-2n} + \frac{1}{(2n-1)64^{2n-1}}, \qquad \zeta(2) = \pi^2/6.$$

For logarithms, rescale by a power of two and use eighty terms of $2 \sum_{j \ge 0} z^{2j+1}/(2j+1)$, where $z = (x-1)/(x+1)$ and the enclosing interval satisfies $0 \le z < 1/2$. After eighty terms its remaining value is between zero and $2z^{161}/(161(1-z^2))$. For positive arctangents, iterate $\arctan x = 2 \arctan(x/(1+\sqrt{1+x^2}))$ until the upper endpoint is at most $1/2$, then use eighty alternating terms and the next-term error. Every normalized angle in (A.1) is checked to lie strictly between zero and $1/2$ before (A.2) is applied.

The portable verifier [14] executes these rational rules and asserts the displayed comparisons. It also checks that the argument of the maximum defining $E(C)$ is positive. Thus its selected branch is certified rather than inferred from a decimal approximation. The smallest certified value of $B_{\mathrm{near}} - D_0$ exceeds $817/10000$, while the weaker table already gives $7/25 - 203/1000 = 77/1000 > 7/100$. No optimizer or loop over m appears in this finite calculation. The passage from these constants to every even $m \ge 6$ is the trigonometric and action argument in Section 7.

Scope

The theorem concerns the original closed braid and its original crossing disks. No terminal tangles are inserted. The crossing powers may have arbitrary signs, but their magnitudes must exceed a threshold depending on the fixed number of syllables. The permutation test for one knot is essential. The proof gives no growing-diagram or moving-color statement, and makes no assertion that these knot types are disjoint from classes that admit another Conway presentation. Its external proof inputs are exactly the three companion results stated in the text.

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